

Wiring the single-relay controller

One ESP32 (WROOM-32U, 38-pin USB-C DevKit) reads the box air and the AC's coil, and one relay swaps the resistance the Frigidaire sees on its room-thermistor connector (CN3). Pin numbers match `controller/cooler-v4.yaml`.

Done so far

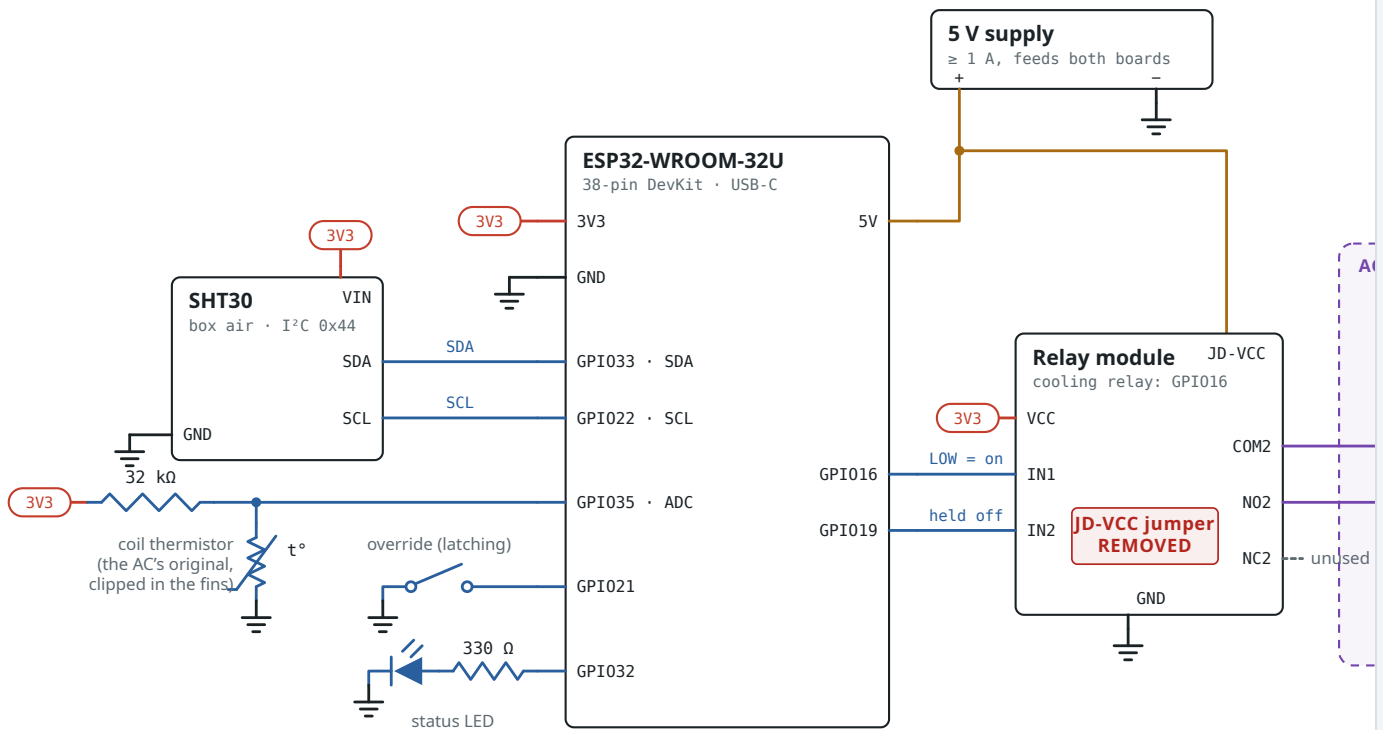
- ✓ **Relays**
Cooling relay on GPIO16, other held off on GPIO19. Both stay off through boot.
- ✓ **Override switch → GPIO21, LED → GPIO32**
Switch changes mode normal ↔ override; state survives a power cycle.
- ✓ **SHT30 on SDA 33 / SCL 22**
Reads at I²C 10 kHz (100 kHz failed on its cable).
- ✓ **Coil thermistor on GPIO35**
32 kΩ pull-up; read 23.4 °C beside the SHT30's 23.4 °C.
- ✓ **AC-side block**
1-2 reads 35 kΩ with the relay open and 13.6 kΩ closed.
- ✓ **MQTT**
TLS to mqtt.example.com as cooler; topics cooler/....
- ✓ **LCD panel (CoolerPanel)**
On the broker with its own login over TLS; settings sent from the panel reach the controller.
- ✓ **Ice-pack bench test (2026-09-28)**
Cooling on/off thresholds, min run and min off timing and defrost trip and recovery all behaved. `fin_cutoff` now defaults to 1 °C.
- ✓ **CoolerApp (Android)**
Built; its own broker login created; APK sent to the phone.

Still to do, in this order

- ① **Set the Frigidaire:** Cool, fan continuous, setpoint at its minimum (≈16 °C)
It must sit between the fake −1 °C (open) and 18 °C (closed).
- ② **Move the thermistor** into the coil fins and connect block 1-2 to CN3
AC unplugged while wiring.
- ③ **Install checklist** in `RULES-v4.md`: AC timing test, then fin calibration

- 4 **Set cooleraset back to 4 °C** for real use
The bench controller is at 12 °C.
- 5 **Check CoolerApp on the phone**
- 6 **Alerts and long-term history** in Node-RED
From the retained cooler/data and cooler/availability.

SCHEMATIC

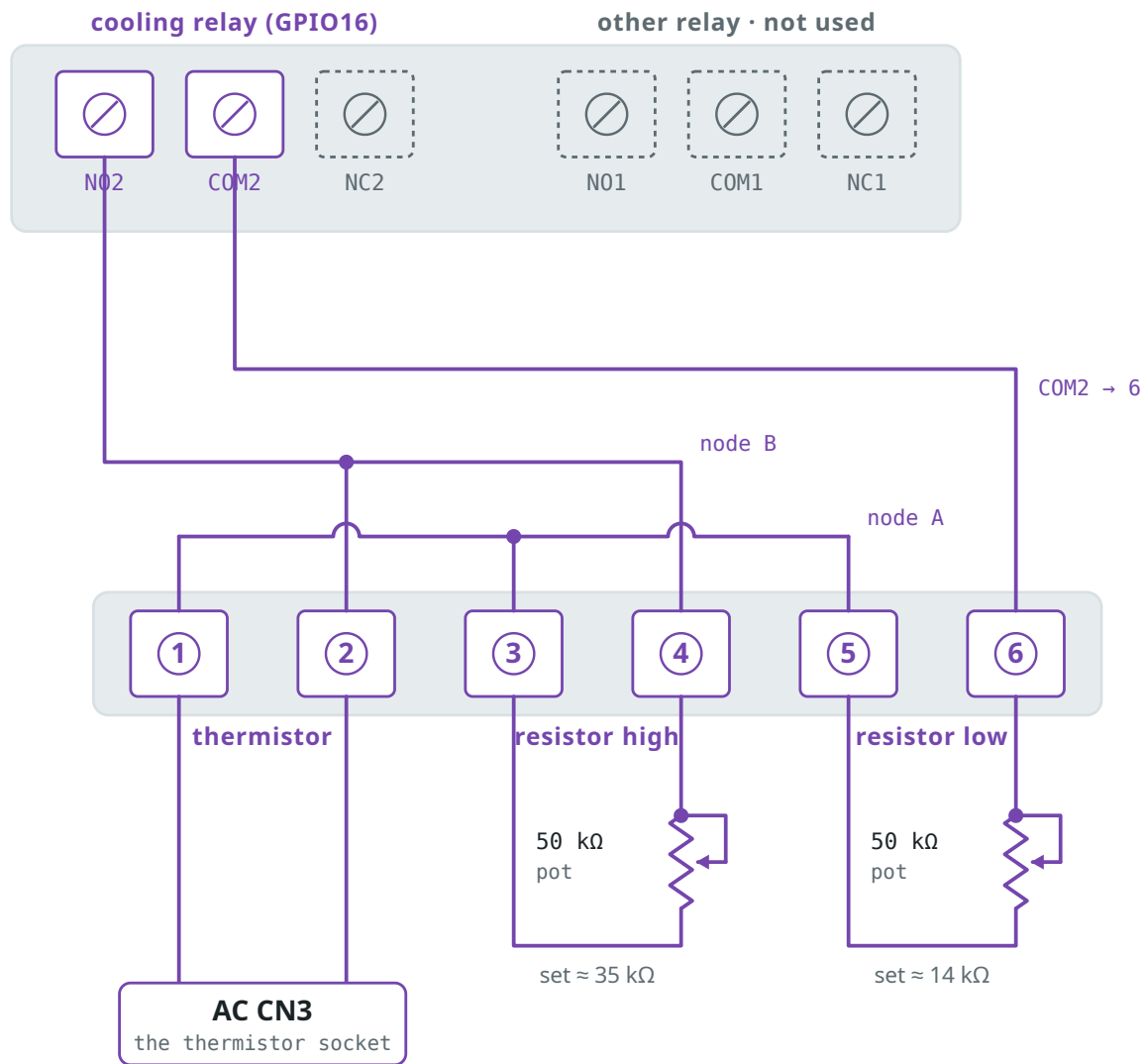


— 3.3 V — 5 V — signal — ground — AC side, isolated

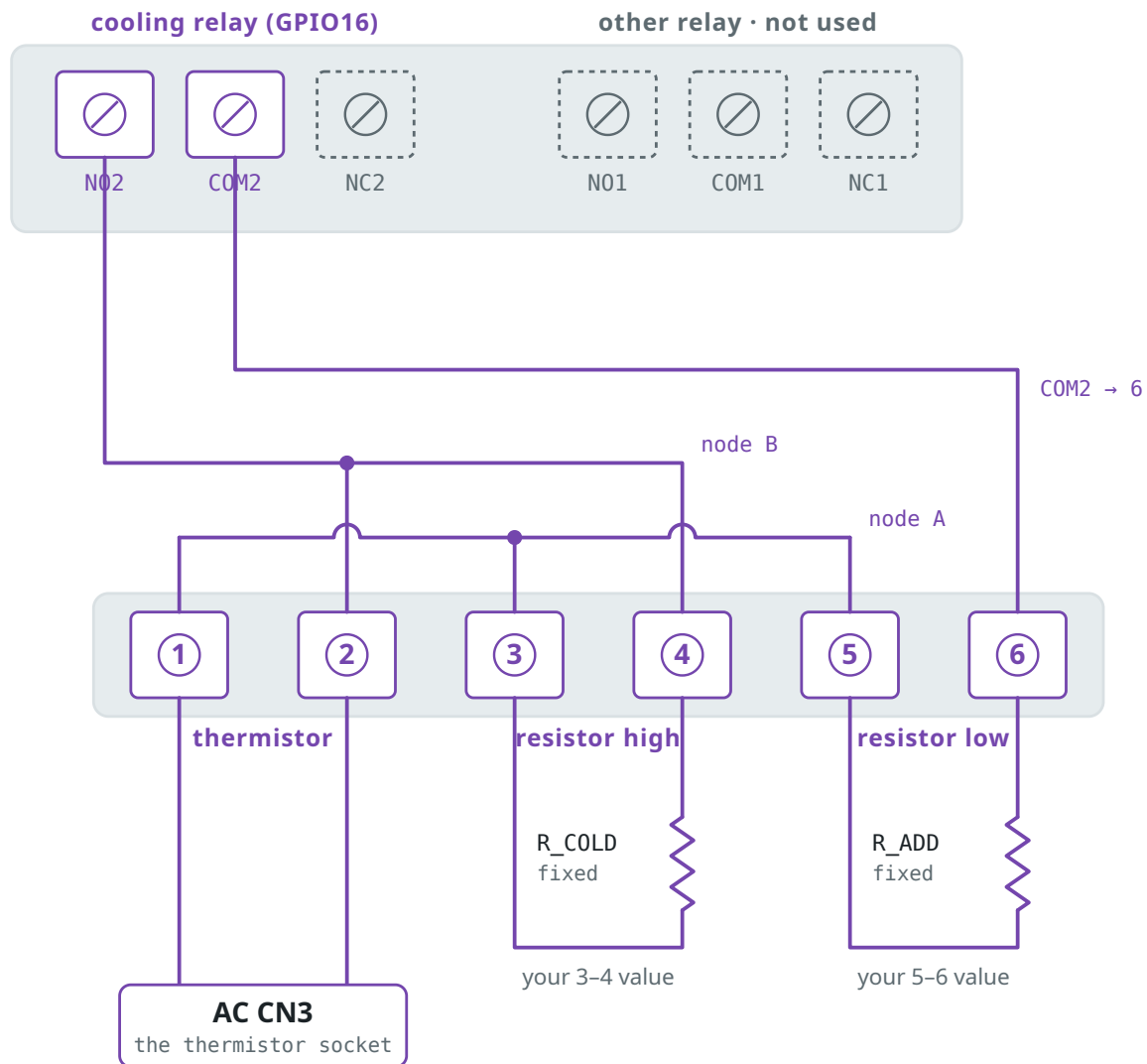
Every **3V3** tag is the same wire as the ESP32's 3V3 pin, and every ground symbol is the same common ground. Wires that cross without a dot are not connected. Relay open: CN3 sees 35 kΩ (≈ -1 °C), the AC reads a cold room and stops. Relay closed: $35\text{ k}\Omega \parallel 22\text{ k}\Omega = 13.6\text{ k}\Omega$ (≈ 18 °C), the AC reads a warm room and cools. Set the AC to its minimum setpoint (≈ 16 °C), between those two.

AC SIDE: RELAY 2 TO THE TERMINAL BLOCK, AS ON YOUR BOARD

STAGE 1 Find the values with pots



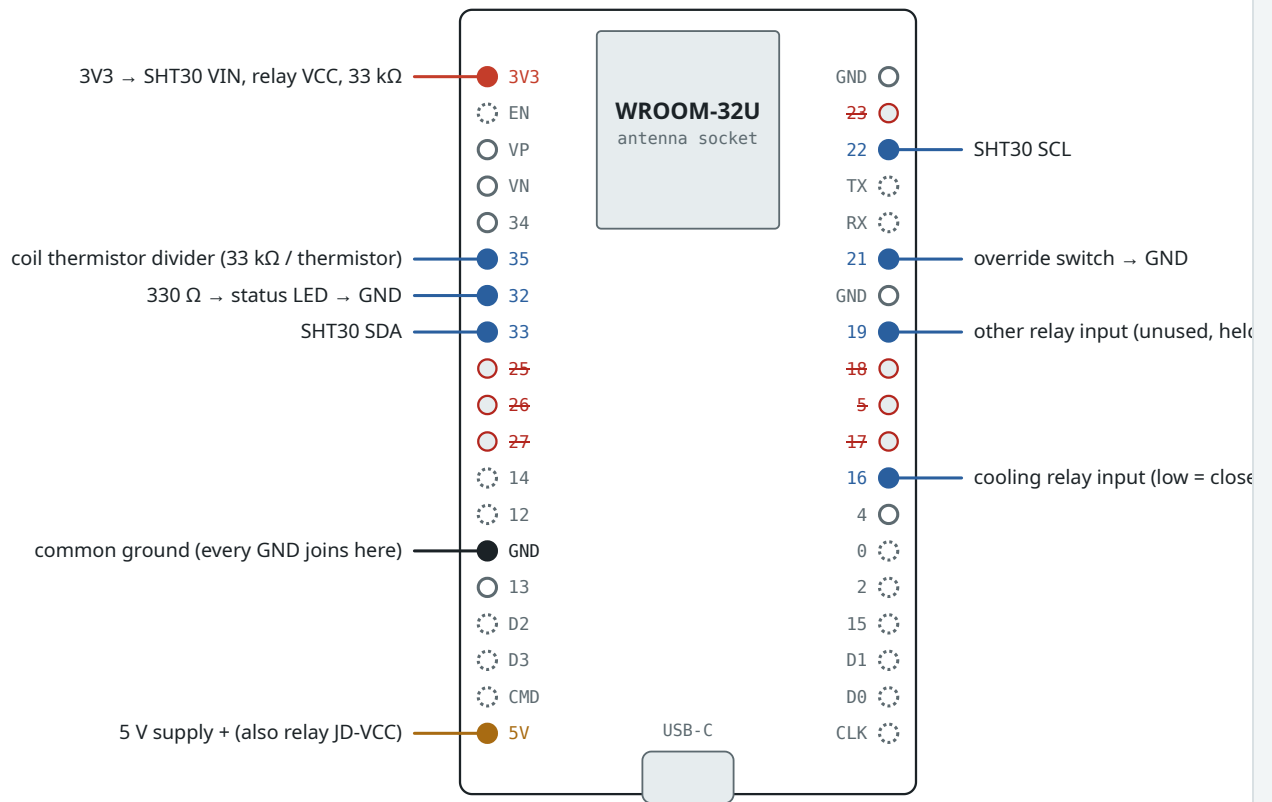
1. **Links** (protoboard traces or short jumpers): 1 – 3 – 5 together (node A); 2 – 4 – NO2 together (node B); COM2 → 6. Leave NC2 and relay 1 empty.
2. A 50 kΩ pot in **3–4** (resistor high) and one in **5–6** (resistor low), each with its wiper tied to one end.
3. AC unplugged and 1–2 disconnected: meter **3–4** and set it to your cold value (≈ 35 kΩ); meter **5–6** and set it to ≈ 14 kΩ. Never plug the AC in with either pot near 0 Ω: CN3 would read as a shorted sensor.
4. Check **1–2**: ≈ 35 kΩ with the relay off, ≈ 13.6 kΩ with it on (as built: 35 k high, 22 k low).
5. Connect 1–2 to CN3 and run the AC. Nudge the pots until it starts reliably with the relay on and stops with it off.
6. AC unplugged, 1–2 disconnected, relay off: meter **3–4** and **5–6** and write both values down.



1. AC unplugged. Take both pots out; the links, relay wires and 1-2 stay as they are.
2. **3-4**: one resistor (R_{COLD}) at the value you wrote down for 3-4. **5-6**: one resistor (R_{ADD}) at the value for 5-6.
3. Nearest precision (1%) value is fine, or two resistors in series to hit an odd value (e.g. 33 k + 2.2 k).
4. Re-check **1-2**: the same readings as stage 1 with the relay off and on. Then reconnect CN3.

Everything in this drawing is on the AC side. None of these positions may connect to the ESP32 or its supply. The coil thermistor goes to the ESP32's GPIO35 divider, not to this block.

HEADER MAP, AS THE BOARD SITS (MODULE UP, USB-C DOWN)



Filled pins are used. **Struck-through red pins (5, 17, 18, 23, 25, 26, 27) are not available on this carrier board: never use them.** Dashed pins are boot-strapping, boot-pulsing or flash pins: leave them free (keep relay wires off 12 and 14). Only one GND pin is needed, since every ground joins on the common ground.

Connections

FROM	TO	NOTE
POWER		
5 V supply +	ESP32 5V	and relay JD-VCC
5 V supply –	ESP32 GND	one common ground for everything
RELAY MODULE		
VCC	ESP32 3V3	jumper removed, so this only powers the optocoupler
IN (cooling)	GPI016	low = closed = cooling requested; switches the COM2/NO2 pair feeding the block
IN (other)	GPI019	unused; the firmware holds it off

FROM	TO	NOTE
JD-VCC	5 V supply +	powers the relay coil
GND	common ground	the GND next to JD-VCC
SENSORS AND CONTROLS		
SHT30 VIN / GND	3V3 / GND	
SHT30 SDA / SCL	GPI033 / GPI022	address 0x44
3V3 → 32 kΩ (22 k + 10 k) → node	GPI035	node → coil thermistor → GND
Override switch	GPI021 ↔ GND	latching; closed = override
LED + 330 Ω	GPI032 → GND	long leg (anode) toward GPIO32
AC SIDE (CN3)		
35 kΩ	CN3 pin 1 ↔ pin 2	always connected
22 kΩ + relay COM2/NO2	CN3 pin 1 ↔ pin 2	in series with each other, in parallel with the 35 kΩ
NC2, other relay	nothing	leave empty

Before you plug in the AC

1. The **JD-VCC jumper** must be off the relay board (done) before VCC goes to 3.3 V.
2. With the AC unplugged, meter the CN3 leads: about **35 kΩ** with the relay open and about **13.6 kΩ** with it closed.
3. Check there is **no continuity** from either CN3 lead to the ESP32's GND or 3V3. The AC board's isolation from mains is unknown, so nothing on that side may touch the ESP32.
4. Keep the relay and the resistor network away from the AC's refrigerant lines. It runs on R32, which is flammable, and the manual warns against sparking switches nearby.

The rest of the bench and install checks are in RULES-v4.md.

Parts

ESP32-WROOM-32U DevKit	38-pin, USB-C; keep the external antenna fitted
SHT30 breakout	box air temperature and humidity
2-channel 5 V relay module	B00XAGT2OG, channel 1 only
AC's original thermistor	moved from CN3 into the coil fins
22 k Ω + 10 k Ω	fin divider pull-up (32 k Ω)
35 k Ω	R_COLD. Not a standard value: use 34.8 k Ω (E96) or 33 k + 2.2 k in series
22 k Ω	R_ADD as built (closed 13.6 k Ω). Use 14 k Ω (6 k + 8 k) instead if the AC's minimum setpoint is above $\approx 17^\circ\text{C}$
330 Ω	LED resistor
LED, latching switch	status, override
5 V supply, ≥ 1 A	feeds the ESP32 and the relay coil

35 k Ω and 14 k Ω come from your measurements of the original thermistor. What matters is the network: aim for your measured cold value open and your warm value closed.